CHANDAN SHAH

Agentic AI Developer • LLM Researcher • Multi-Agent Systems
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PROFESSIONAL SUMMARY

Computer Engineering student at IOE Pulchowk Campus (TU) with hands-on experience building autonomous multi-agent systems, RAG pipelines, and LLM-powered applications using LangChain, LangGraph, FastAPI, and PyTorch. Self-directed researcher who has re-implemented DeepSeek MLA and MoE architectures from scratch and built a GPT-2 (124M) from first principles. Seeking a remote AI/ML internship where I can contribute to production-grade systems.

TECHNICAL SKILLS

Languages: Python (Advanced), C, C++, JavaScript, SQL, Bash
AI/ML Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn
Agentic AI & LLM: LangChain, LangGraph, LangSmith, ReAct Agents, Multi-Agent Systems, Tool Calling
LLM Research: RLHF, GRPO, DPO, SFT, Chain-of-Thought, Inference Optimization, LLM Scaling Laws
Generative AI & RAG: OpenAI GPT-4, Anthropic Claude API, RAG, Fine-Tuning, Embedding Models, Semantic Search
Vector Databases: Pinecone, ChromaDB, FAISS, Knowledge Graphs, Document Chunking
NLP & Deep Learning: Transformer Architectures, CNNs, RNNs, Attention Mechanisms, Vision Transformers (ViT)
Backend & APIs: FastAPI, RESTful APIs, Webhooks, n8n Automation, Streamlit, Event-Driven Architecture
Cloud & DevOps: Docker, Git/GitHub, Linux, PostgreSQL, MongoDB, AWS/GCP (Fundamentals)

PROJECTS

Autonomous Multi-Agent Research & Execution System | github.com/chandan11248 *Jun 2025 – Nov 2025*
Tech Stack: *LangChain · LangGraph · n8n · OpenAI GPT-4 · FastAPI · Python*

- Designed a ReAct-based autonomous AI agent with multi-step task planning, dynamic tool selection, and persistent cross-session memory — replacing a fully manual research workflow
- Implemented parallel multi-agent collaboration in LangGraph using conditional branching, dynamic routing, and supervisor agent patterns for complex task delegation across specialized sub-agents
- Integrated 5+ external tools (web scraping, REST APIs, data summarization) and deployed orchestration on n8n with retry logic, error handling, and real-time monitoring

Enterprise RAG Chatbot with Intelligent Document Querying *Sep 2024 – Dec 2024*
Tech Stack: *LangChain · Pinecone · ChromaDB · FastAPI · OpenAI Embeddings · Python*

- Built a RAG system supporting natural language queries over 1,000+ private documents using OpenAI embedding models and vector databases for semantic retrieval
- Engineered multi-document prompt chaining and context window management to enable complex cross-document reasoning with measurably reduced hallucination rates
- Exposed the pipeline as a FastAPI service with webhook support, enabling integration into downstream enterprise workflows

AI-Powered Job Hunting Agent (Telegram Bot) | github.com/chandan11248/JOB-HUNTING-AGENT *Oct 2025 – Nov 2025*
Tech Stack: *LangGraph · Python · Telegram Bot API · Jooble API · Remotive API · PDF Generation*

- Built an end-to-end autonomous job application bot on Telegram using LangGraph orchestration — automating job discovery across Jooble, Remotive, and LinkedIn simultaneously
- Implemented AI-driven resume customization pipeline that tailors resume content per job description and auto-generates submission-ready PDFs

GPT-2 Transformer Implementation from Scratch *Jun 2025 – Oct 2025*

Tech Stack: *Python · PyTorch · NumPy*

- Implemented the full GPT-2 architecture (124M parameters) from first principles — multi-head self-attention, positional encoding, layer normalization, and autoregressive decoding
- Validated model perplexity on benchmark text datasets, achieving performance within 8% of the reference GPT-2 implementation

DeepSeek Architecture Re-implementation *2025*

Tech Stack: *Python · PyTorch · NumPy · CUDA · Hugging Face*

- Studied 5+ DeepSeek research papers (V2, V3, R1, MoE, Coder), focusing on Multi-head Latent Attention (MLA), Mixture-of-Experts (MoE), and GRPO training
- Re-implemented core DeepSeek components — MLA, sparse MoE routing, and auxiliary-loss-free load balancing — in PyTorch to validate theoretical understanding

Deep Learning Foundations — CNNs, RNNs & Transformers | github.com/chandan11248/deep-learning *2024 – Present*

Tech Stack: *Python · PyTorch · Keras · TensorFlow · Jupyter Notebook*

- Built and trained CNNs, RNNs, and Transformer architectures from scratch as part of structured self-directed research into modern deep learning
- Exploring Vision Transformers (ViT) and attention mechanisms in computer vision as an ongoing research thread

Generative AI & LangChain Mastery Repository | github.com/chandan11248/GENERATIVE-AI-LEARNIG *2024 – Present*

Tech Stack: *LangChain · OpenAI · RAG · Pinecone · Prompt Engineering · Jupyter Notebook*

- Comprehensive hands-on repository covering LLM integration, RAG pipelines, prompt engineering, vector databases, and agentic workflows with real-world mini-projects

EDUCATION

Bachelor of Engineering — Computer Engineering *Expected 2028*

Tribhuvan University, IOE Pulchowk Campus, Kathmandu, Nepal

Relevant Coursework: Artificial Intelligence, Machine Learning, NLP, Data Science, DBMS, Data Structures & Algorithms

CERTIFICATIONS

Claude with the Anthropic API — Anthropic *Mar 2026*

Certificate No: 976s9i5kfuz7 https://verify.skilljar.com/c/976s9i5kfuz7

Advanced Deep Learning with Keras — DataCamp *Oct 2025*

Certificate No: Certificate #43,662,780

Supervised Learning with scikit-learn — DataCamp *Sep 2025*

Certificate No: Certificate #41,513,364

RESEARCH & OPEN SOURCE

- Active research: DeepSeek paper series (V2/V3/R1/MoE), LLM post-training (RLHF, GRPO, DPO), reasoning mechanisms, inference-time scaling, Vision Transformers (ViT)
- 18 public GitHub repositories spanning deep learning, agentic AI, generative AI, and LangChain/LangGraph workflows
- Open for: AI Agent Development, Automation Projects, Research Collaborations, Remote Internships, Open-Source Contributions